the input image contributes a scaled and shifted version of the point spread function to the output image. As viewed from the output side, each pixel in the output image is influenced by a group of pixels from the input signal. For one-dimensional signals, this region of influence is the impulse response flipped *left-for-right*. For image signals, it is the PSF flipped *left-for-right* and *top-for-bottom*. Since most of the PSFs used in DSP are symmetrical around the vertical and horizonal axes, these flips do nothing and can be ignored. Later in this chapter we will look at nonsymmetrical PSFs that must have the flips taken into account.

Figure 24-3 shows several common PSFs. In (a), the **pillbox** has a circular top and straight sides. For example, if the lens of a camera is not properly focused, each point in the image will be projected to a circular spot on the image sensor (look back at Fig. 23-2 and consider the effect of moving the projection screen toward or away from the lens). In other words, the pillbox is the point spread function of an out-of-focus lens.

The **Gaussian**, shown in (b), is the PSF of imaging systems limited by *random* imperfections. For instance, the image from a telescope is blurred by atmospheric turbulence, causing each point of light to become a Gaussian in the final image. Image sensors, such as the CCD and retina, are often limited by the scattering of light and/or electrons. The Central Limit Theorem dictates that a Gaussian blur results from these types of random processes.

The pillbox and Gaussian are used in image processing the same as the *moving average filter* is used with one-dimensional signals. An image convolved with these PSFs will appear blurry and have less defined edges, but will be lower in random noise. These are called **smoothing filters**, for their action in the time domain, or **low-pass filters**, for how they treat the frequency domain. The **square** PSF, shown in (c), can also be used as a smoothing filter, but it is not circularly symmetric. This results in the blurring being different in the diagonal directions compared to the vertical and horizontal. This may or may not be important, depending on the use.

The opposite of a smoothing filter is an **edge enhancement** or **high-pass filter**. The spectral inversion technique, discussed in Chapter 14, is used to change between the two. As illustrated in (d), an edge enhancement filter kernel is formed by taking the *negative* of a smoothing filter, and adding a delta function in the center. The image processing which occurs in the retina is an example of this type of filter.

Figure (e) shows the two-dimensional sinc function. One-dimensional signal processing uses the windowed-sinc to separate frequency bands. Since images do not have their information encoded in the frequency domain, the sinc function is seldom used as an imaging filter kernel, although it does find use in some theoretical problems. The sinc function can be hard to use because its tails decrease very slowly in amplitude ($1/x$), meaning it must be treated as infinitely wide. In comparison, the Gaussian's tails decrease very rapidly (e^{-x^2}) and can eventually be truncated with no ill effect.